

TECHNICAL REPORT · ROAD TO GREEN

Rooftop Solar PV System
Design Basis & Cost Estimate

Marriott Hotel – Mezzanine Roof – Phase 1

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Client	Kampala Marriott Hotel
System capacity	185.4 kWp DC / 160 kW AC, grid-tied

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Document Control

Rev	Date	Description	Prepared	Checked	Approved
A	3/11/2026	Version A	Mahil Fernando	Prasanna Dissanayake	Manjula Perera

Reference documents

Reference	Title
D18102-DD-A-104	Mezzanine floor plan incl. roof PV zones (architect)
PV-LAY-D18102-002	PV layout with ballast structure, 10° tilt
EPJ-26110401 V1.0	Mounting structure drawings (2P and 1P tables)

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1. Executive Summary

This report sets out the design basis, layout, energy yield and budget cost for a grid-tied rooftop solar photovoltaic (PV) system on the mezzanine-level roof of the Marriott Hotel,

The system has no battery storage and supplies the hotel's own load behind the utility meter.

185.4 kWp	291 MWh	USD 103k	4.9 yrs
DC capacity (299 × 620 Wp)	Year-1 generation	Total project cost (0.55 USD/Wp)	Simple payback

Key findings

From Energy audit and feasibility study key findings.

Item	Value
Roof area (mezzanine roof)	1,150 m² (Available area for solar installation)
Plant capacity	185.4 kWp
Modules	299 × 620 Wp (13 rows of 23 solar modules)
Structure	Fixed-tilt ballast structure, 10°
Inverters	2 × 80 kW (160 kW AC)
Hotel consumption	1,000,000 kWh/month
Self-consumption	100% (no export, no battery)
Year-1 generation	291 MWh (2.4% of annual consumption) – PVsyst P50
Avoided tariff	USD 0.07/kWh
Year-1 savings	USD 20,380
Total project cost	USD 102,789
Simple payback	4.9 years
Project IRR	21.0%
NPV (25 years at 10%)	USD 104,491
Cost of solar energy (LCOE)	0.040 USD/kWh
Road to 500 kWp	315 kWp, needs other roofs or car-park canopies

Table 1: Summary of Energy Audit Finding & Recommended PV Solar Installation

Recommendation

Phase 1: Proceed with the 185.4 kWp mezzanine roof system.

Phase 2: Proceed with 250.4 kWp the car park

2. Site and Design Basis

2.1 Site and load

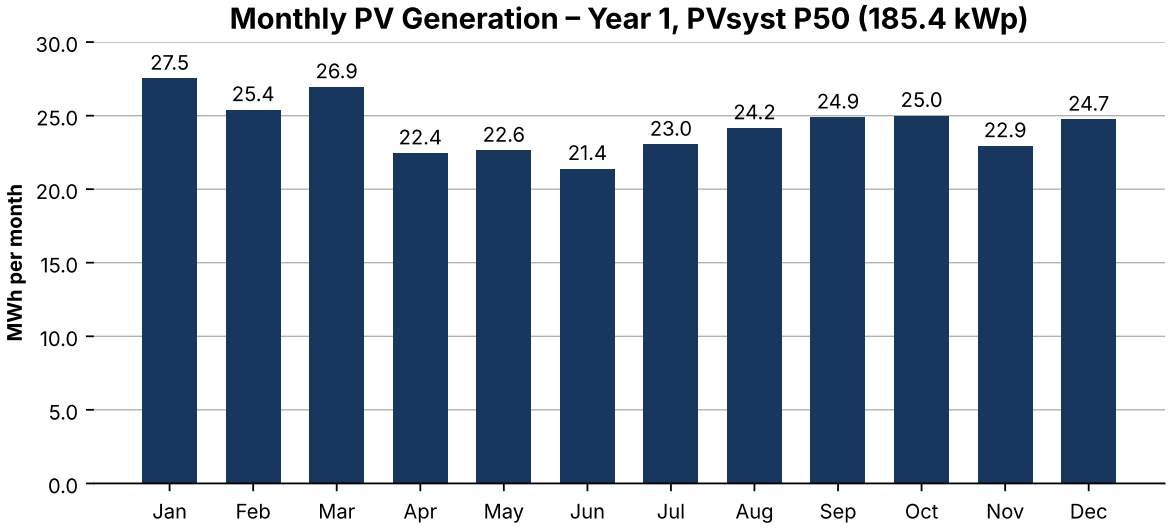
Facility	Marriott Hotel –PV on the mezzanine-level roof
Location	0°18'11.34"N, 32°35'13.76"E
Roof	Flat reinforced-concrete slab, single level, with parapet.
Annual electricity consumption	12,000,000 kWh
Average monthly energy consumption	1,000,000 kWh
Average / daytime demand	
Grid connection	Behind-the-meter, LV main switchboard
Energy storage	None



Figure 1: PV module installation on the mezzanine roof (AI Generated)

3. Estimation of PV Solar Generation

- Year-1 generation 291,051 kWh, on average 24,254 kWh/month and 797 kWh/day
- 25-year generation 6,871 MWh*
- share of hotel uses 100 %, 2.4 % of annual consumption



*1.0 % degradation in year 1, then 0.4 %/yr

Source: PVsyst V8.1.8 simulation (P50, Meteonorm 9.0 weather, performance ratio 85.4 %, specific yield 1,570 kWh/kWp) – Annex 7. Output is highest from January to March and lowest in June.
The best month is about 29 % above the lowest. The hotel load is higher than the PV output in every hour, so all this energy is used on site. Figures are P50 as simulated, before shading and soiling losses (P90 ≈ 8 % lower).

4. Solar Impact on Time-of-Use Tariff

The hotel is billed on a time-of-use (TOU) tariff, so each kWh is worth a different amount depending on when it is used.

Tariff period	Hours	Indicative rate (USD/kWh)
Off-peak	00:00–06:00	0.054
Shoulder	06:00–18:00	0.070
Peak	18:00–24:00	0.086

Table 2:Bands per uganda ERA TOU structure; rates indicative (USD/kWh)

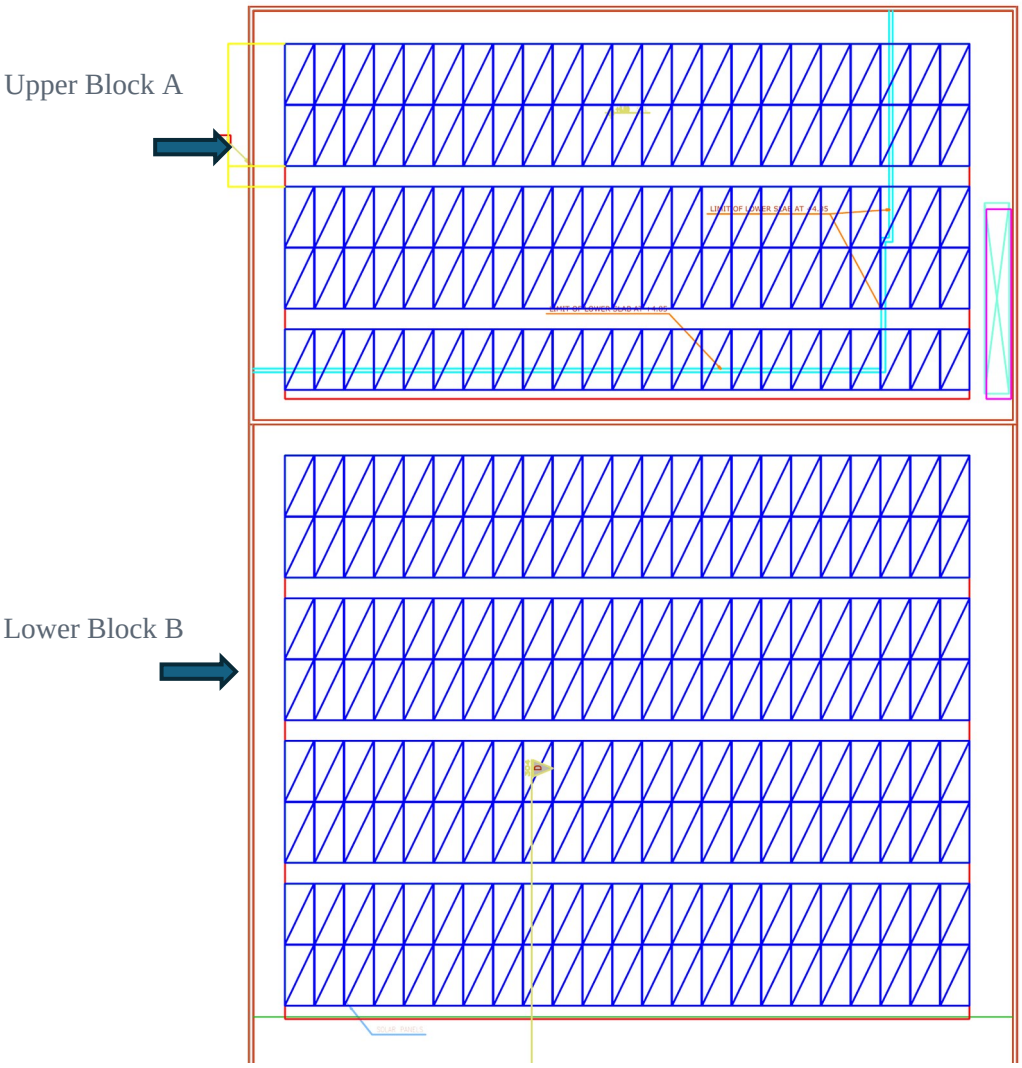
Period	Hotel use (kWh/yr)	Solar (MWh/yr)	Share of solar	Grid after solar (kWh/yr)	Reduction	Cost before (USD/yr)	Cost after (USD/yr)	Saving (USD/yr)
Off-peak	2,090	0.0	0.0%	2,090	0.00%	113,648	113,648	0
Shoulder	6,954	291.1	100.0%	6,663	4.19%	486,965	466,585	20,380
Peak	2,956	0.0	0.0%	2,956	0.00%	254,079	254,079	0
Total	**12,000**	**291.1**	**100.0%**	**11,709**	**2.43%**	**854,692**	**834,312**	**20,380**

Table 3 : Energy reduction percentage & cost saving per year

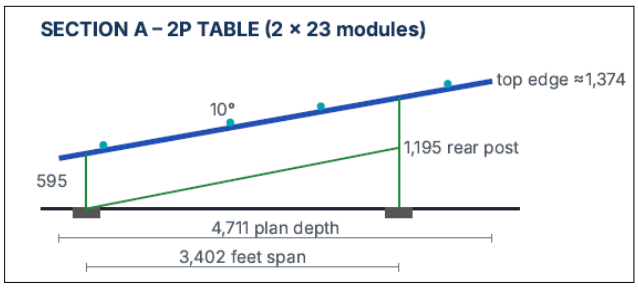
*Forecasted Energy demand for year assumed by september energy bill*12 All solar output (including the small output just after 18:00) is counted in the shoulder period.

5. PV Array Layout

Each row holds 23 modules on Enerack R80 rails 26,630 mm long, supported by 11 frames per table at 2,600 mm centers. The upper block carries two 2P tables and one 1P table; the lower block carries four 2P tables.



Zone	Tables	Rows	Modules	kWp
Upper block A	2 × 2P + 1 × 1P	5	115	71.3
Lower block B	4 × 2P	8	184	114.1
Total	**7 tables**	**13**	**299**	**185.4**



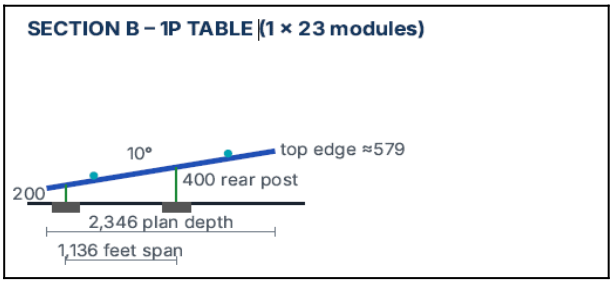


Figure 4 : PV Layout and Table Structure details,

6. Electrical Design

6.1 Configuration

Item	Inverter INV-01	Inverter INV-02	Total
Rated AC power	80 kW	80 kW	160 kW
Strings	8 × 19 modules	3 × 19 + 5 × 18 modules	16 strings
Modules	152	147	299
DC capacity	94.2 kWp	91.1 kWp	185.4 kWp
DC/AC ratio	1.18	1.14	1.16

Table 5: System Configuration & Inverter Capacity of each block

6.2 Single-line arrangement

1. PV strings: 4 mm² double-insulated PV cable through the cable trays, via a DC isolator and a Type II DC SPD for every string, to the inverters on the roof in a shaded, ventilated location.
2. Inverters (400 V, 3-phase) → AC combiner panel with 2 × 160 A MCCB incomers, Type II AC SPD and 400 A roof isolator.
3. AC combiner panel → 4C × 240 mm² Cu XLPE/SWA → existing Masterpact NW10 H1 (1000 A) solar incomer on the main LV switchboard.
4. Earthing: panel-to-panel bonding of the module frames, connected to the building earth and integrated with the lightning protection system.

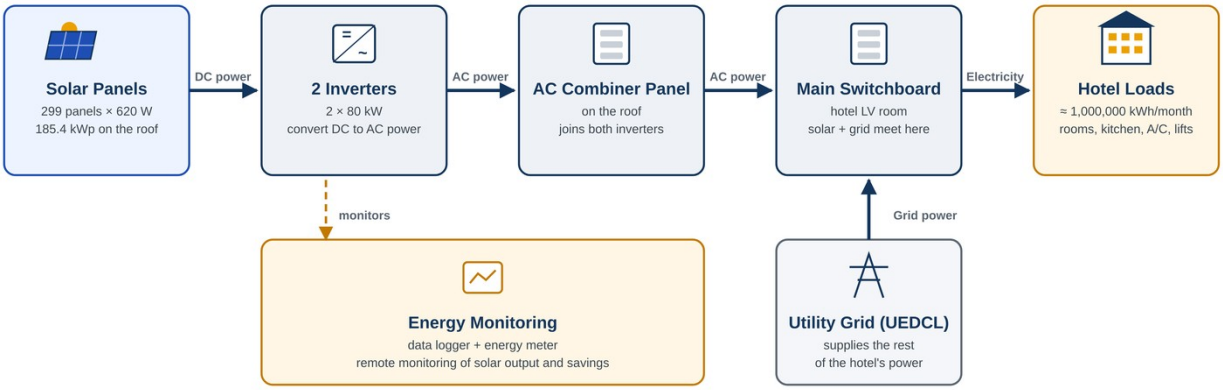


Figure 5 : Roof Top PV System Block Diagram

Component	Description	Size / Quantity
Solar panels	Generate DC electricity	299 × 620 W = 185.4 kWp
Inverters	Convert DC power to AC power	2 × 80 kW
AC combiner panel	Joins both inverter outputs on the roof	1 panel, 400 A
Main switchboard	Where solar and grid power meet (hotel LV room)	Existing, 1000 A
Utility grid (UEDCL)	Supplies the rest of the hotel's electricity	Existing supply
Expected generation	Solar energy used by the hotel each year	291,000 kWh/yr (24,250 kWh/month)

Table 6 each item and in the block and Size/Quantity

7. Cost Estimate (Bill of Quantities)

No.	Description	Qty	Rate (USD)	Amount (USD)
1	Supply of PV modules, 620 Wp (299 Nos Modules)	185.4 kWp	0.185	34,378
2	Supply of String inverters, 80 kW	2 nos	4,000	8,000
3	Supply of Mounting structure per Wp	185.4 kWp	0.095	17,634
4	Supply of DC system	Lot	USD 0.029/Wp	5,376
5	Supply of AC system	Lot	Lump sum	6,245
6	Earthing, equipotential bonding & lightning protection integration	Lot	Lump sum	1,847
7	Installation, testing & commissioning	Lot	Lump sum	15,279
8	Transport, logistics & customs clearing	Lot	Lump sum	8,650
	Sub-total			97,409
	Contingency		7 %	**5,380**
	Total Project Value (excluding taxes)			**102,789**
	Specific cost			0.554 USD/Wp

Table 7: Project Cost Estimation

8. Financial Analysis

The solar system needs a total investment of USD 102,789 (total project value, excluding taxes). At an avoided grid tariff of USD 0.07/kWh, rising 2% a year, it saves about USD 20.4k in its first year (PVsyst P50 generation).
The investment pays back in about 4.9 years. Over 25 years the project gives an IRR of 21.0% and an NPV of USD 104,491 at a 10% discount rate, with total savings of about USD 614,000.

Solar energy costs about USD 0.040/kWh, below the grid tariff, so every unit generated reduces the hotel's electricity cost. No O&M or inverter-replacement cost is included in the cash flow.

Total investment (excluding taxes)	USD 102,789
Avoided grid tariff (year 1) / escalation	USD 0.07/kWh / 2 % per year
Year-1 savings	USD 20,380
Simple payback	4.9 years
Project IRR (25 years, unlevered)	21.0 %
NPV at 10 % discount rate	USD 104,491
Levelized cost of energy (LCOE)	USD 0.040/kWh
25-year cumulative savings	USD 613,840

Table 8 : Financial Analysis

Tariff sensitivity

The project remains viable across a wide tariff range. Each USD 0.01/kWh change in the avoided tariff moves the NPV by about USD 29.6k and the payback by about 0.6 years. The break-even tariff (NPV = 0 at a 10 % discount rate) is about USD 0.035/kWh, roughly 50 % below the current USD 0.07/kWh, so the investment stays profitable even if the effective tariff falls significantly.

Avoided tariff (USD/kWh)	Payback (yrs)	IRR	NPV 10 % (USD)
0.070	4.9	21.0%	104,424
0.080	4.3	23.9%	134,026
0.090	3.9	26.8%	163,628
0.100	3.5	29.6%	193,230
0.120	2.9	35.3%	252,433

Table 9: Tariff Sensitivity

Annual cash flow (selected years, USD)

For an investment of USD 102,789, the system saves about USD 20.4k in its first year, rising with the tariff*. The investment is recovered in about 4.9 years, and over 25 years the hotel's cumulative net cash flow reaches USD 511,051.

Year	Generation (kWh)	Tariff (USD/kWh)	Savings	Net cash flow	Cumulative
0				(102,789)	(102,789)
1	291,051	0.070	20,380	20,380	(82,409)
2	286,988	0.071	20,498	20,498	(61,911)
3	285,840	0.073	20,824	20,824	(41,087)
4	284,697	0.074	21,155	21,155	(19,932)
5	283,558	0.076	21,492	21,492	1,561
6	**282,424**	**0.077**	**21,834**	**21,834**	**23,395**
7	**281,294**	**0.079**	**22,182**	**22,182**	**45,577**
8	**280,169**	**0.080**	**22,535**	**22,535**	**68,112**
9	**279,048**	**0.082**	**22,894**	**22,894**	**91,006**
10	**277,932**	**0.084**	**23,258**	**23,258**	**114,264**
12	**275,713**	**0.087**	**24,005**	**24,005**	**161,898**
15	**272,418**	**0.092**	**25,170**	**25,170**	**236,229**
18	**269,162**	**0.098**	**26,391**	**26,391**	**314,168**
20	**267,013**	**0.102**	**27,238**	**27,238**	**368,217**
25	**261,715**	**0.113**	**29,476**	**29,476**	**511,051**

Table 10: Annual Cash Flow

**Grid price saved for each kWh of solar. Starts at USD 0.07 and rises 2% a year to about USD 0.113 by year 25.*

9. Expansion Options

The target of about 500 kWp exceeds what the architect's solar zone on the mezzanine roof can hold. Because the hotel load is around ten times the PV output, every additional kWp would also be fully self-consumed, so expansion can be added in stages without a battery.

Option	Capacity	Year-1 energy	Budget cost (USD)	Remarks
Phase 1 – mezzanine roof (this report)	185.4 kWp	291 MWh	102,789	Ready for detailed design
Phase 2 – car-park solar canopies	250 kWp	370 MWh	187,500	Adds shaded parking; steel canopy cost
Indicative total	435 kWp	661 MWh	290,289	5.5 % of hotel consumption

10. Assumptions, Exclusions and Next Steps

Key assumptions

- The hotel load is about 1,000,000 kWh per month with a continuous daytime demand well above 160 kW.
- The avoided tariff is USD 0.07/kWh escalating at 2 %/yr, to be confirmed from the latest utility bills.
- Prices are 2026 East African market levels in USD, excluding taxes.
- Energy yield from the PVsyst V8.1.8 simulation, P50 as simulated (291,051 kWh/yr); shading and soiling are not yet modelled and a P90 year is about 8 % lower.

Exclusions

- Battery storage, diesel-generator synchronisation and export metering.
- Roof waterproofing renewal, structural strengthening, and civil works to the LV room.
- Utility network upgrades, permit fees and financing costs.

Annexes

1.Single Line Diagram
2.Layout Drawing
3.Data Sheets of Inverter
4.Data Sheet of Solar Module
5.Stuctrural Drawing
6.Quotation
7.PVsyst Simulation Report